



Chinook Pedicab Fabrication Manual

Vancouver People Mover Tours - Hand-Made Fabrication Method

1. Introduction

The Chinook is a unique, human-powered pedicab with electric assist, designed for high-performance at a low fabrication cost. This manual focuses on the artisanal "Blue Board" lamination method using bamboo and carbon fiber.

2. The "Blue Board" Template System (Jigs)

Fabrication relies on high-density XPS "Blue Board" jigs (moulds) to shape the curved bamboo components. These templates are typically cut from 18mm plywood, MDF, or high-density foam.

Component ID	Description	Template Length	Function
BB-01	Main Chassis Rail (L/R)	2400 mm (94.5")	Primary structural curve of the frame.
BB-02	Wheel Arch Support	1200 mm (47.2")	Rear axle mounting and wheel clearance.
BB-03	Passenger Bench Frame	1300 mm (51.2")	Support structure for the seat.
BB-04	Steering Column Strut	900 mm (35.4")	Connection for the front fork/steering.

3. Materials & Lamination Schedule

The frame is constructed by gluing thin bamboo strips into the jigs. Carbon fiber is integrated into the core for high tensile strength.

- **Bamboo Strips:** 5mm thickness layers.
- **Adhesive:** High-strength industrial epoxy.
- **Reinforcement:** Internal Carbon Fibre core layers.
- **Clamping:** Industrial heat-shrink tubing is used to provide uniform pressure across organic curves.

Section	Bamboo Layers	CF Reinforcement	Total Thickness
Main Chassis Rails	12 Layers	2 Layers (Internal)	65 mm
Main Joints	10 Layers	4 Layers (Wrap)	60 mm
Wheel Arches	8 Layers	2 Layers (Outer)	45 mm

4. Fabrication Workflow

1. **Template Cutting:** Cut two of each BB template to allow for parallel lamination.
2. **Molding:** Apply epoxy to bamboo strips and nest them into the BB-01/BB-02 jigs.
3. **Compression:** Slide industrial heat-shrink tubing over the wetted assembly and apply heat for uniform clamping force.
4. **Curing:** Allow 24-36 hours for a full structural cure before removing from the jig.
5. **Assembly:** Interface the 5-bolt hub system with the carbon-reinforced bamboo axle.

5. Technical Specifications

- **Payload Capacity:** 300kg+ (Pilot + 2 Adults + Cargo).
- **Drive System:** 500W Electric Assist motor compatibility.
- **Braking:** Hydraulic disk brakes.
- **Chassis:** Bamboo-monocoque for impact absorption.

Manual Version 3.0 - Verified for Vancouver People Mover Tours Project.

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